UNIVERSITY OF CHITTAGONG
Department of Computer Science and Engineering

Assignment

Course Title : Database Management System
Course Code No. : CSE 414

Submitted to:
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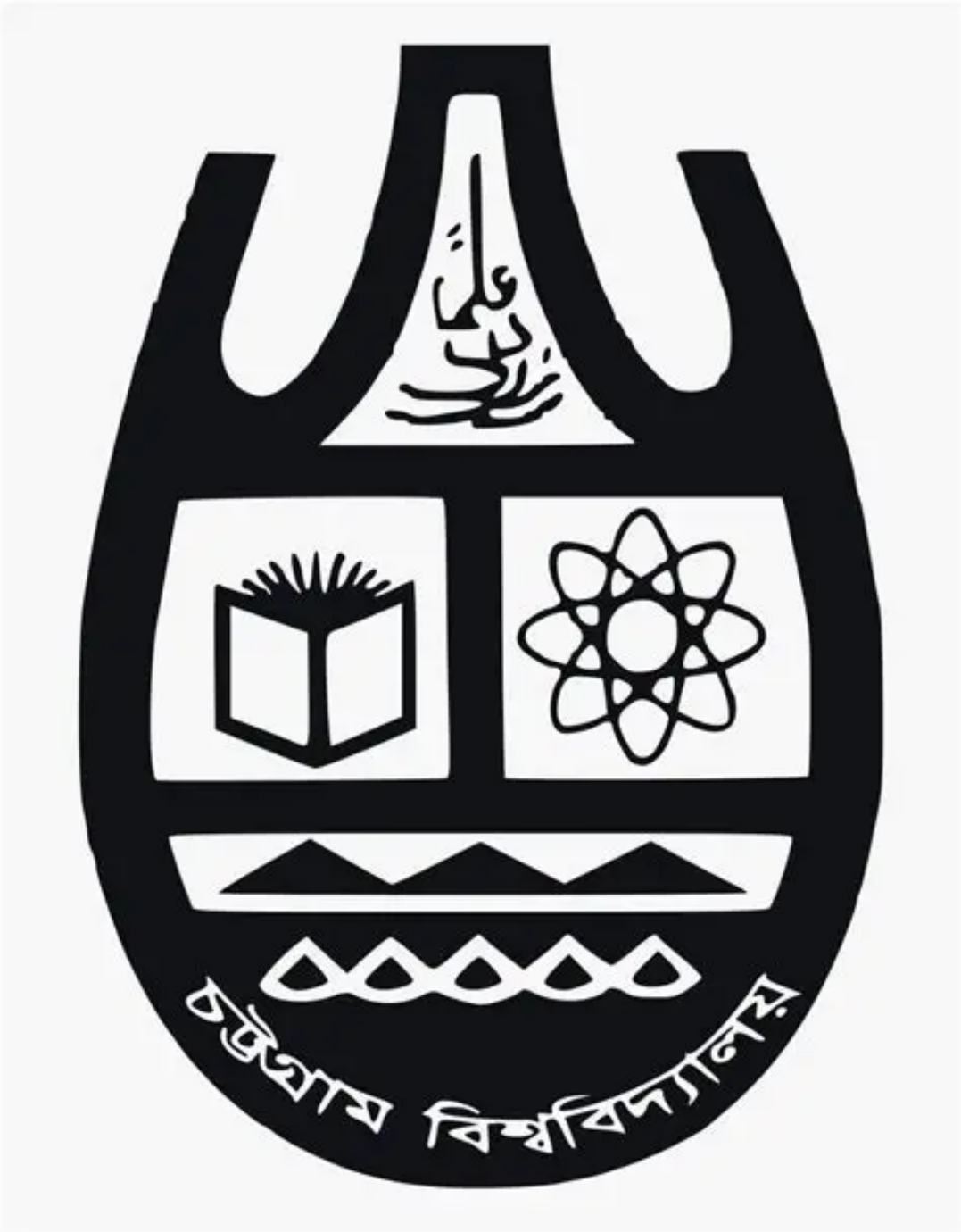
Chapter 6

Question 1
Analyze, solve, and present the solution of Problem 6.1.

Answer:

For the car insurance company, the main entity sets are **Customer**, **Car**, **Accident**, **Policy**, and **Payment**.



E-R diagram for Problem 6.1

Entities:

- Customer (<u>customer_id</u>, name, address, phone)
- Car (<u>car_id</u>, license_no, make, model, year)
- Accident (<u>accident_id</u>, accident_date, location, description)
- Policy (<u>policy_no</u>, issue_date, type, status)
- Payment (<u>payment_id</u>, period_from, period_to, due_date, received_date, amount)

Relationships:

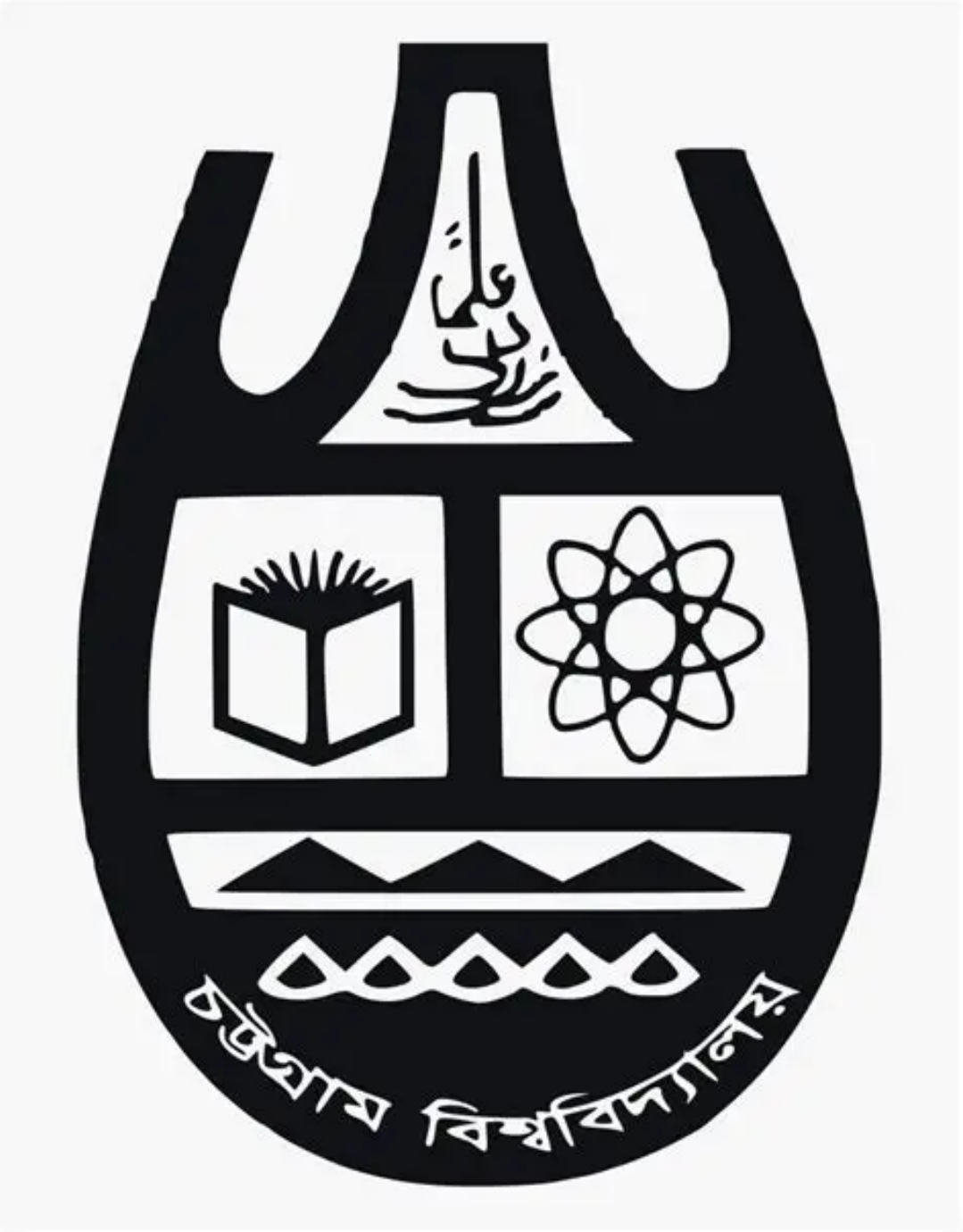
- Owns(Customer, Car): one customer can own many cars
- Has_Accident(Car, Accident): one car can have many accidents
- Covers(Policy, Car): many-to-many
- Has_Payment(Policy, Payment): one policy has many payments

Question 2

Analyze, solve, and present the solution of Problem 6.2.

Answer:

(a) Using a ternary relationship with Exam as an entity



E-R diagram for Problem 6.2(a)

The required entity sets are:

- Student (student_id, name)
- Course (course_id, title)
- Section (section_id, sec_id, semester, year)
- Exam (exam_id, exam_type, exam_date, max_marks)

A ternary relationship **Marks(Student, Section, Exam)** is used with attribute **score**. This is appropriate because a student's mark depends on the student, the section, and the exam together.

(b) Using only a binary relationship between Student and Section

E-R diagram for Problem 6.2(b)

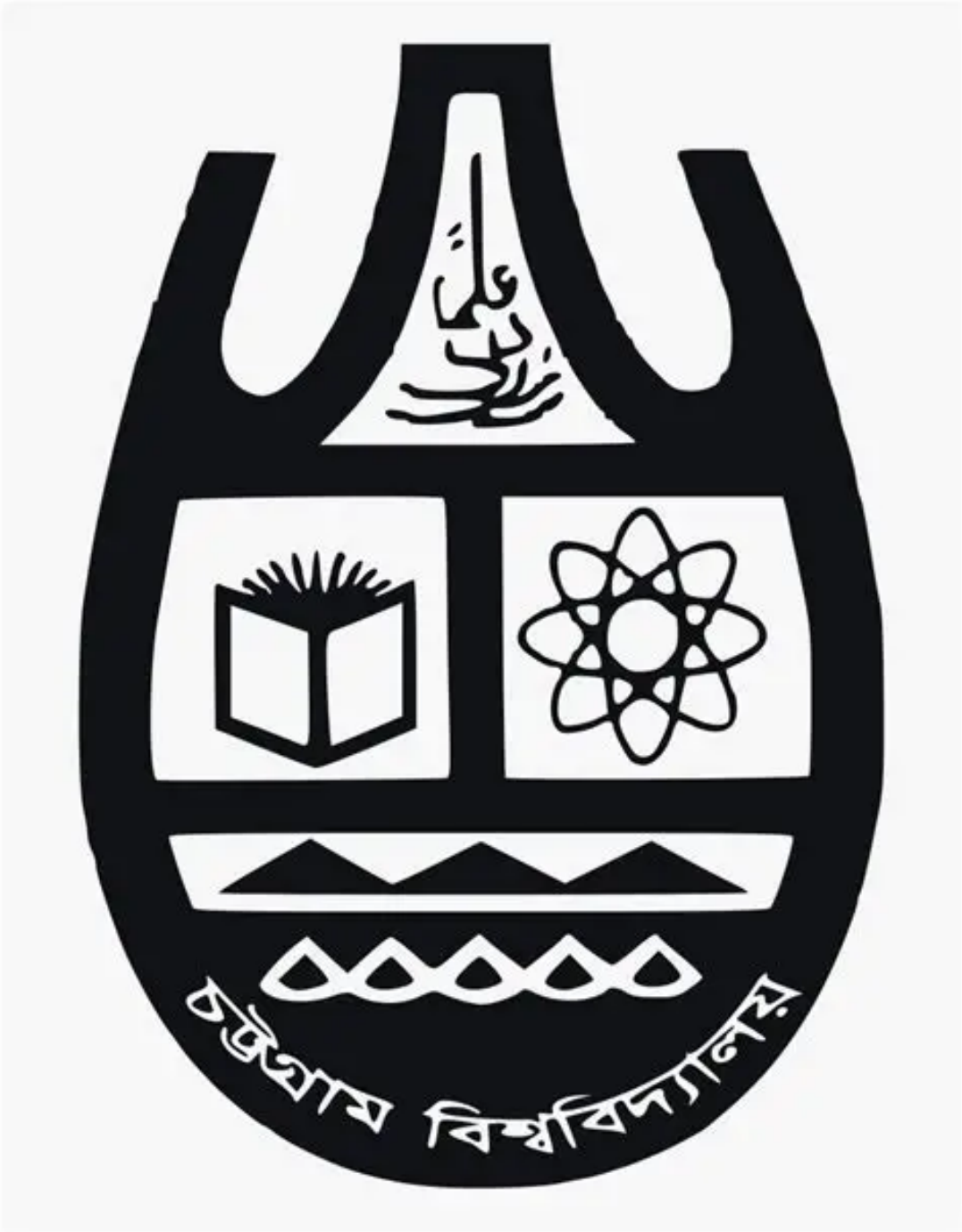
A suitable design is to introduce:

- Enrollment (enrollment_id, student_id, section_id)
- Exam (exam_id, section_id, exam_type, exam_date, max_marks)
- Exam_Result (enrollment_id, exam_id, score)

This design keeps one binary student-section relationship while allowing multiple exam results.

Question 3
Analyze, solve, and present the solution of Problem 6.3.

Answer:



E-R diagram for Problem 6.3

For the sports team scoring system, the main entity sets are:

- Team (team_id, name, city)
- Player (player_id, name, position, jersey_no)
- Match (match_id, match_date, venue, opponent, team_score, opponent_score)

A relationship **Participation(Player, Match)** stores match-specific statistics such as:

- minutes_played
- started_flag
- goals
- assists
- penalty_goals
- own_goals

Derived values such as total goals, total assists, and total wins should be computed from the stored data.

Question 4
Analyze, solve, and present the solution of Problem 6.5.

Answer:

(a) Disconnected graph: A disconnected E-R graph means that two or more parts of the design are not linked by any relationship. This may indicate either independent subdomains or a missing relationship.

(b) Cycle in a graph: A cycle means that by traversing relationships, it is possible to return to the same entity set. This is not necessarily an error; it often reflects valid semantic connections.

Question 5
Analyze, solve, and present the solution of Problem 6.7.

Answer:

A weak entity should not always be converted into a strong entity because doing so introduces redundancy and hides existence dependence.

Example:

- Employee (emp_id, name)
- Dependent is a weak entity identified through Employee

If Dependent is converted into a strong entity as Dependent(emp_id, dependent_name, age), then emp_id must be repeated for every dependent.

Therefore, weak entities are useful because they clearly represent:

- existence dependence
- identification through the owner
- cleaner conceptual modeling

Question 6
Analyze, solve, and present the solution of Problem 6.9.

Answer:

Suppose the relation is: advisor(s_ID, i_ID)

To make the advisor relationship one-to-one:

- each student must have at most one instructor
- each instructor must advise at most one student

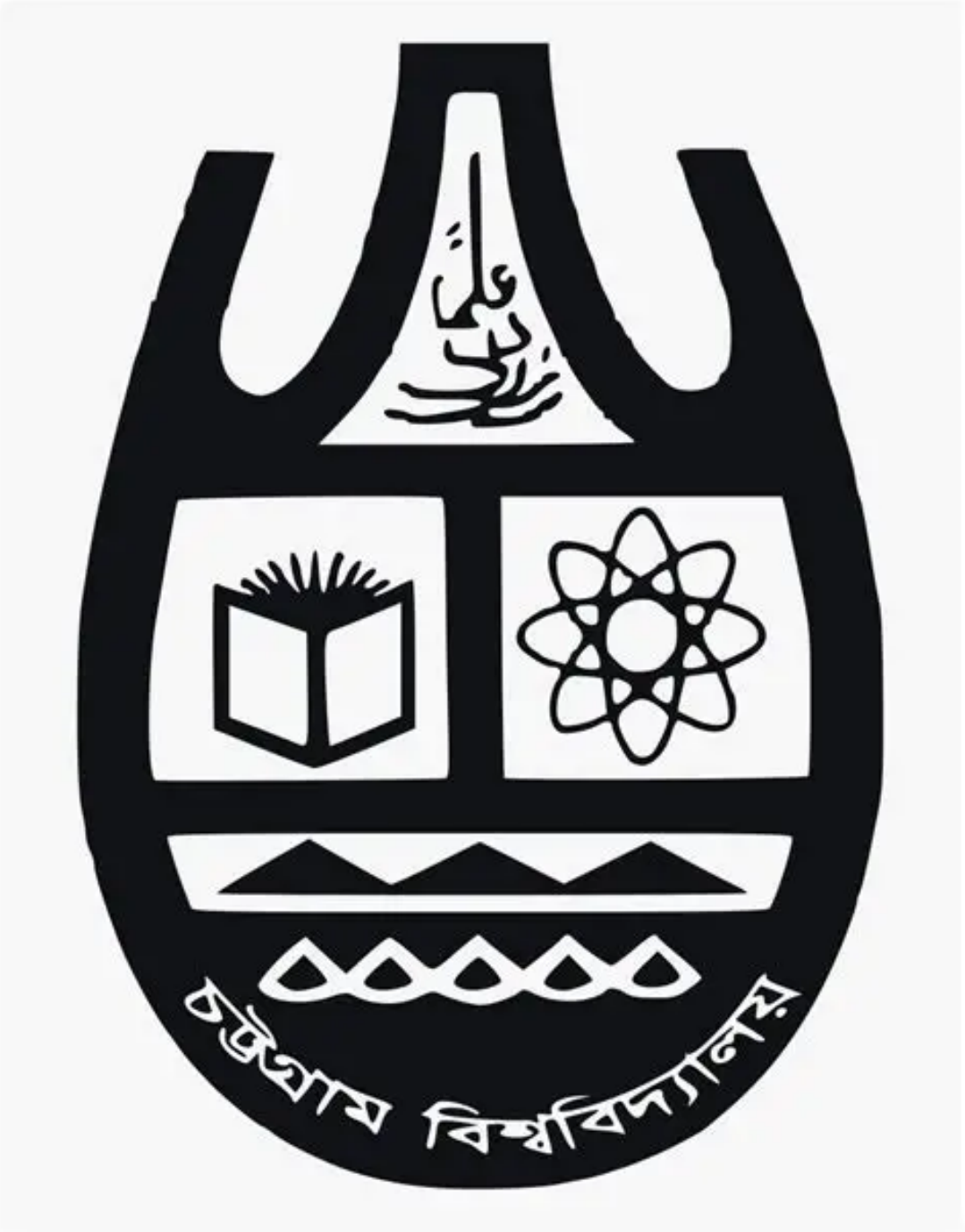
So:

- s_ID must be unique
- i_ID must also be unique

A suitable relational form is: advisor(s_ID PRIMARY KEY, i_ID UNIQUE)

Question 7
Analyze, solve, and present the solution of Problem 6.13*.

Answer:



Temporal E-R design for Problem 6.13

To represent temporal information, valid time should be associated with entities, attributes, and relationship sets.

Temporal design:

- Student(student_id, entity_valid_time)
- Instructor(instructor_id, entity_valid_time)
- Advisor(student_id, instructor_id, valid_time)

Temporal attributes may be represented separately, such as:

- Student_Name(student_id, name, valid_time)
- Student_TotCred(student_id, tot_cred, valid_time)
- Instructor_Name(instructor_id, name, valid_time)
- Instructor_Salary(instructor_id, salary, valid_time)

Reduction to relations:

- Student(student_id, entity_valid_from, entity_valid_to)
- Instructor(instructor_id, entity_valid_from, entity_valid_to)
- Student_Name(student_id, valid_from, name, valid_to)
- Student_TotCred(student_id, valid_from, tot_cred, valid_to)
- Instructor_Name(instructor_id, valid_from, name, valid_to)
- Instructor_Salary(instructor_id, valid_from, salary, valid_to)
- Advisor(student_id, instructor_id, valid_from, valid_to)

Question 8
Analyze, solve, and present the solution of Problem 6.14.

Answer:

Superkey: any set of attributes that uniquely identifies a tuple.

Candidate key: a minimal superkey.

Primary key: the candidate key selected as the main identifier.

Example: If Student(ID, email, name) has both ID and email unique, then:

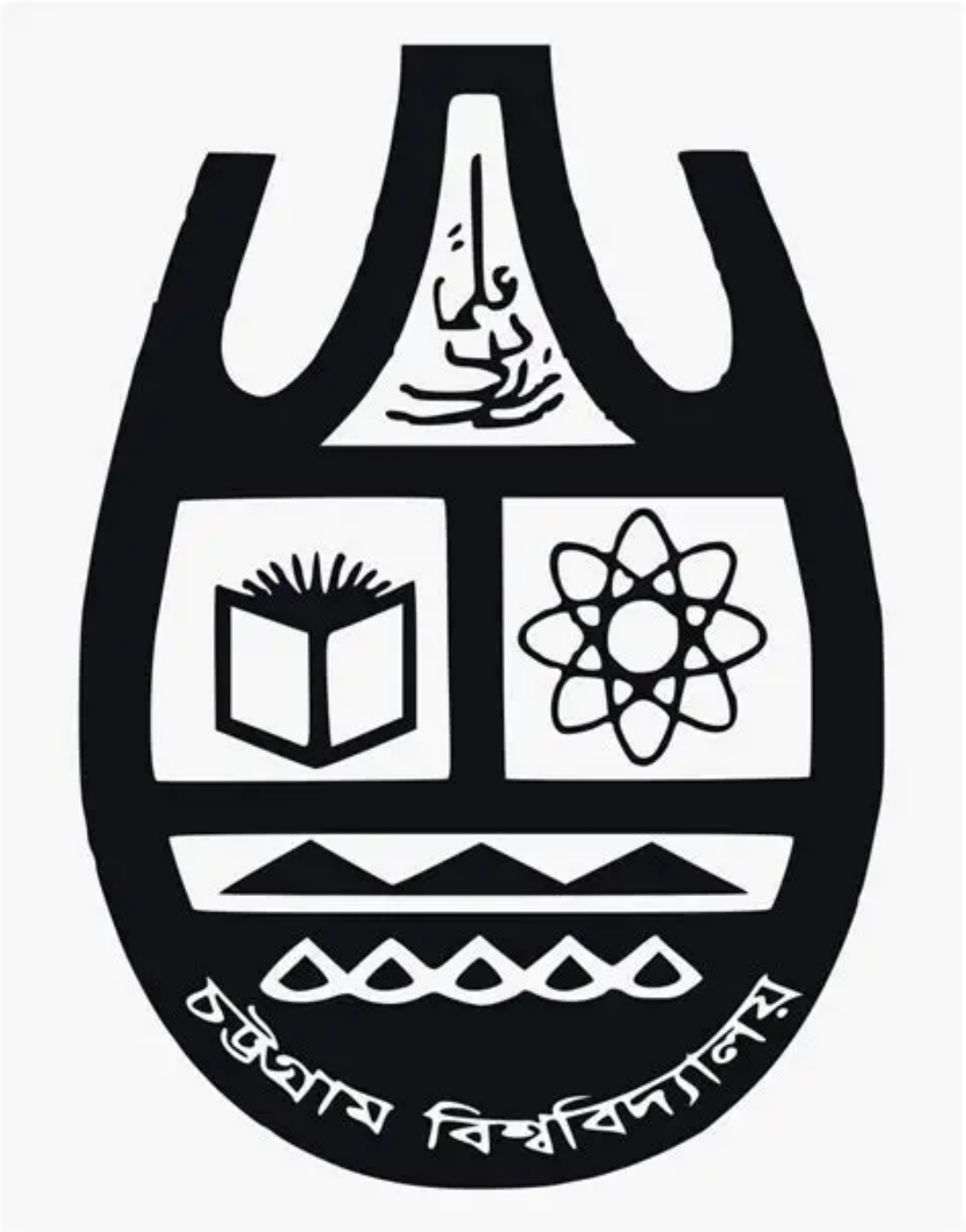
- ID is a candidate key
- email is also a candidate key
- (ID, email) is a superkey but not a candidate key

If ID is chosen as the main identifier, then it is the primary key.

Question 9

Analyze, solve, and present the solution of Problem 6.15.

Answer:



E-R diagram for Problem 6.15

The hospital database may be modeled using:

- Patient (patient_id, name, dob, sex, address, phone, blood_group)

- Doctor (doctor_id, name, specialization, phone, department)
- Test_Exam (log_id, test_type, test_date, result_summary, notes)

Relationships:

- For_Patient(Test_Exam, Patient)
- Conducted_By(Test_Exam, Doctor)

This model is clean because each test or examination is treated as an entity.

Question 10
Analyze, solve, and present the solution of Problem 6.19.

Answer:

Weak entities remain useful even when they can be transformed into strong entities because they naturally express:

- existence dependence
- identification relative to an owner entity
- better conceptual clarity
- avoidance of artificial keys

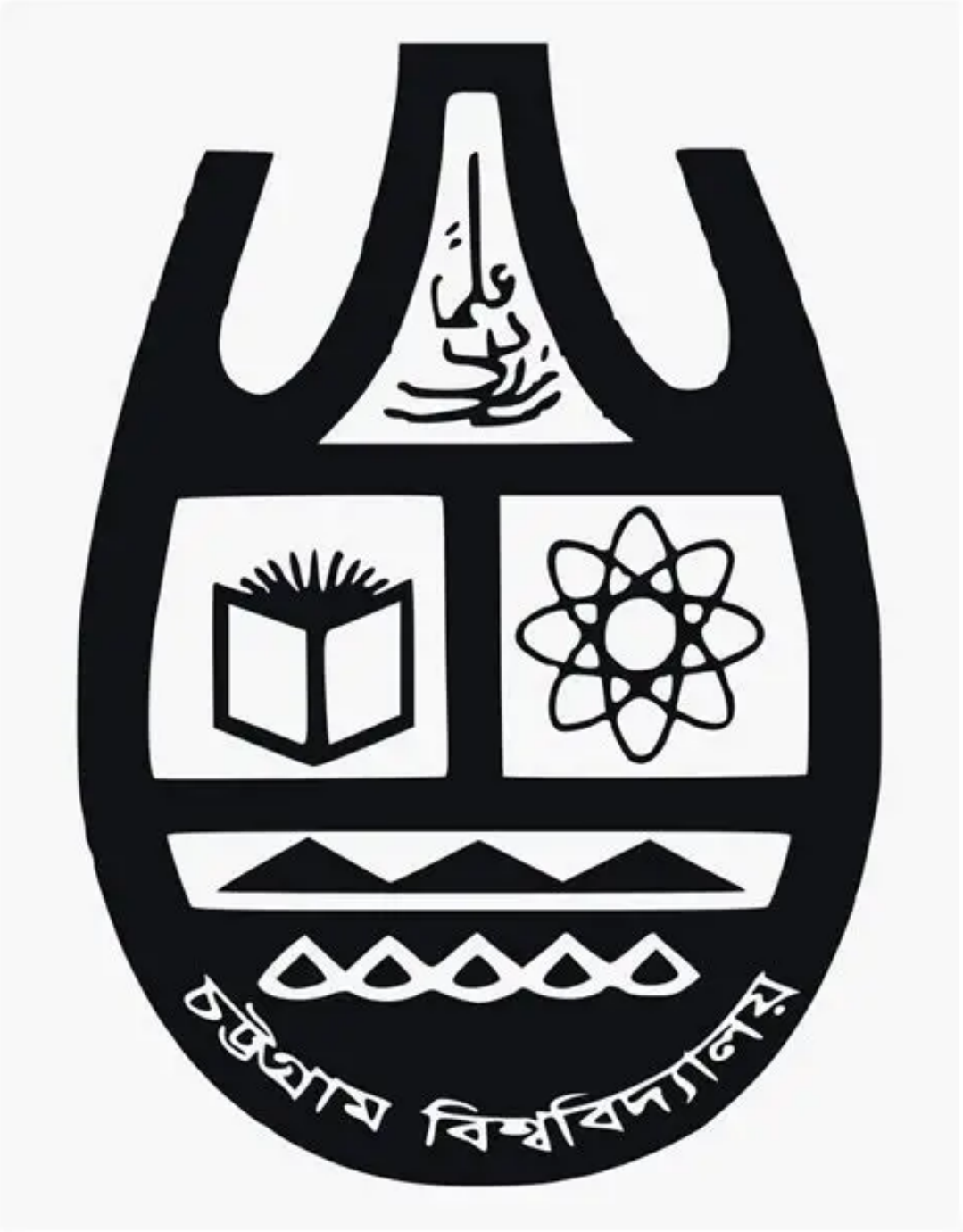
Therefore, weak entities are important for semantic modeling.

Question 11

Analyze, solve, and present the solution of Problem 6.21.

Answer:

(a) Add Blu-ray discs and downloadable video



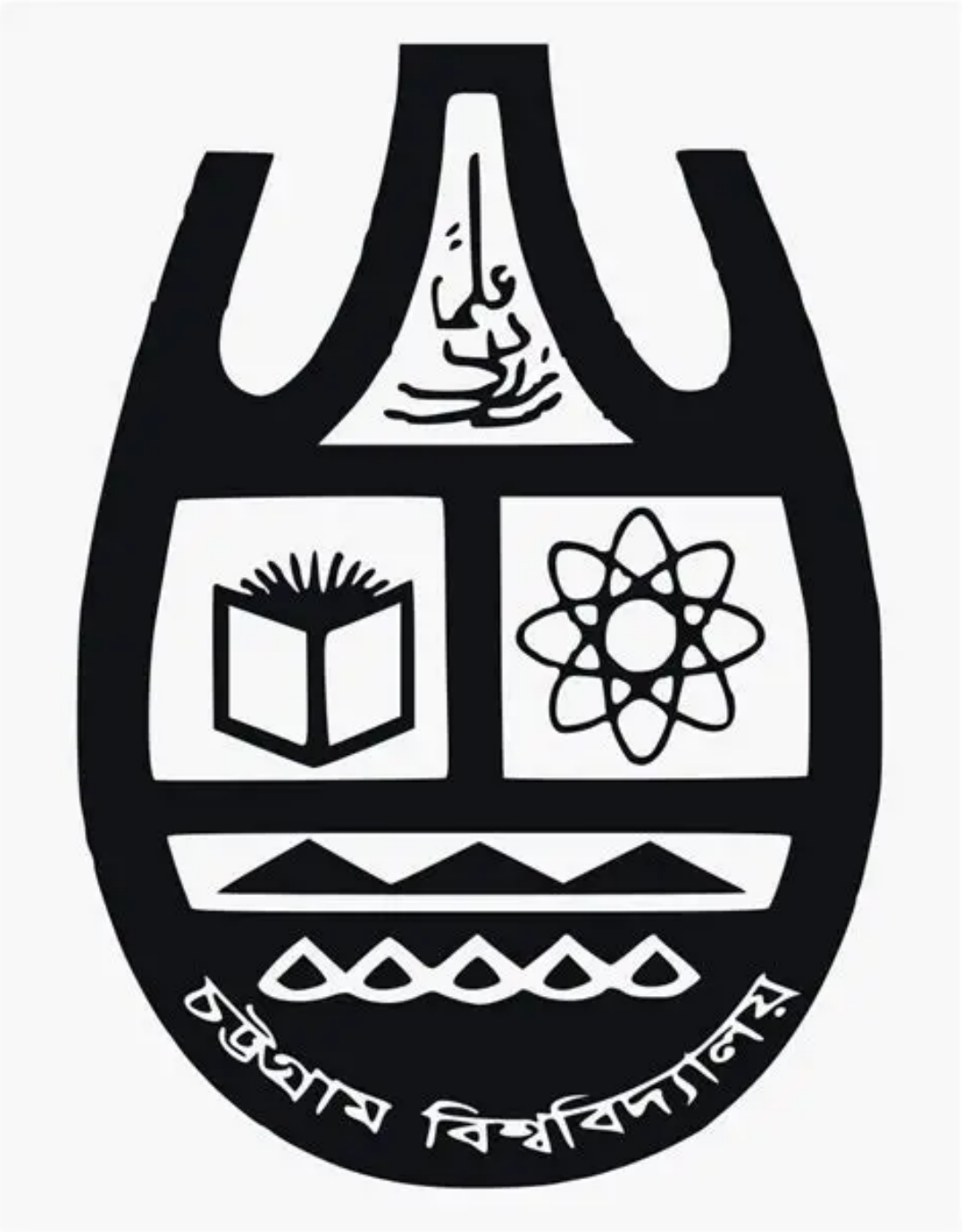
Extended E-R diagram for Problem 6.21(a)

A suitable extension is:

- Video_Work(video_id, title, year, category)
- BluRay_Video(video_id, price, region_code)
- Download_Video(video_id, price, file_format, drm_type)

This allows the same content to exist in one or both formats with separate prices.

(b) Allow shopping basket to contain books and video items



Extended E-R diagram for Problem 6.21(b)

Introduce:

- Store_Item(item_id, title)

Subclasses:

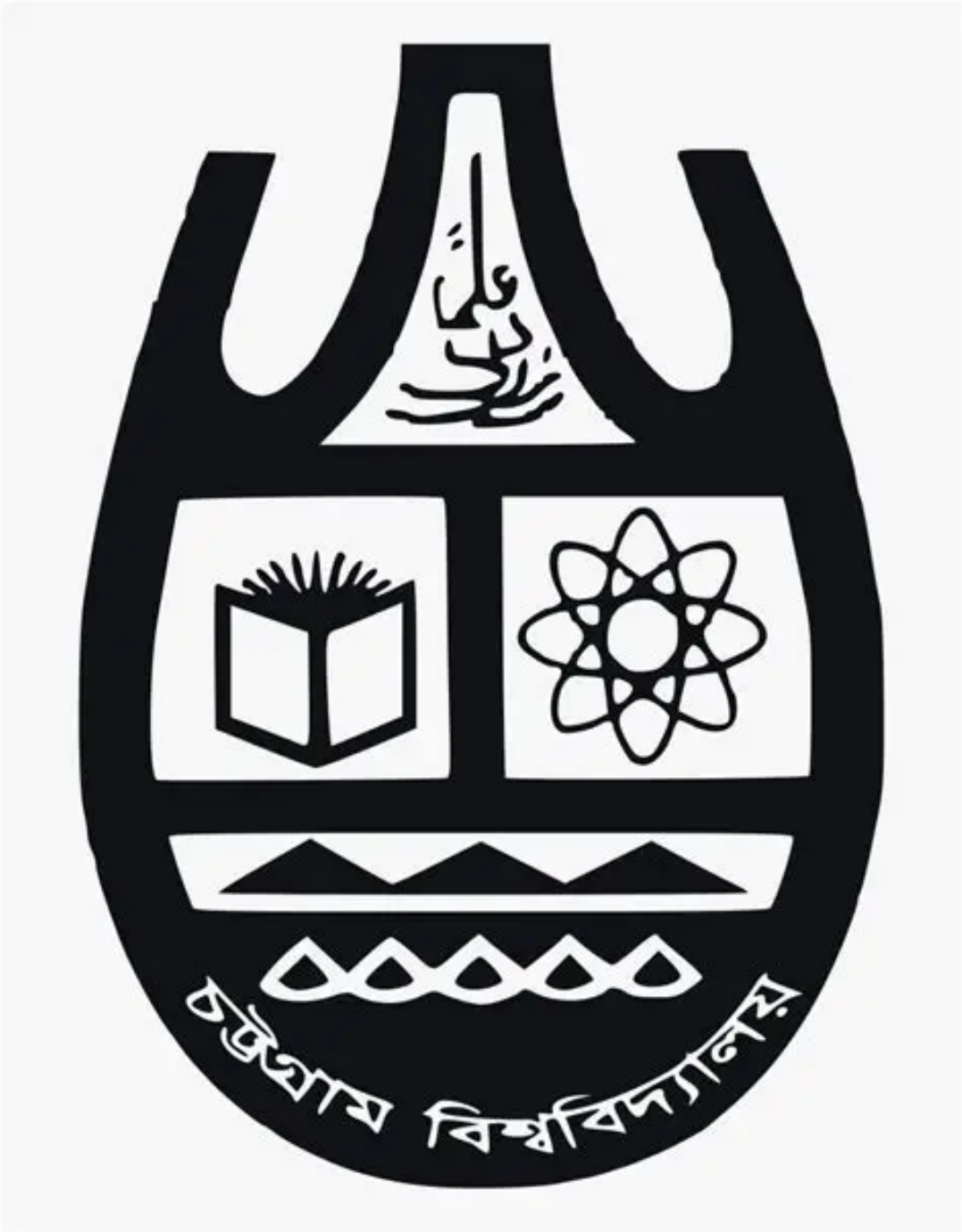
- Book(item_id, ISBN, year, price)
- BluRay(item_id, price)
- DownloadVideo(item_id, price)

Then replace: contains(shopping_basket, book)

with: contains(shopping_basket, store_item, quantity)

Question 12
Analyze, solve, and present the solution of Problem 6.22 (ER diagram only).

Answer:



E-R diagram for Problem 6.22

A suitable design includes:

- Brand(brand_id, brand_name)
- Model(model_id, brand_id, model_name, model_year, body_style, base_price)
- Option(option_id, option_name, description, option_price)
- Vehicle(VIN, model_id, color, manufacture_date, status)

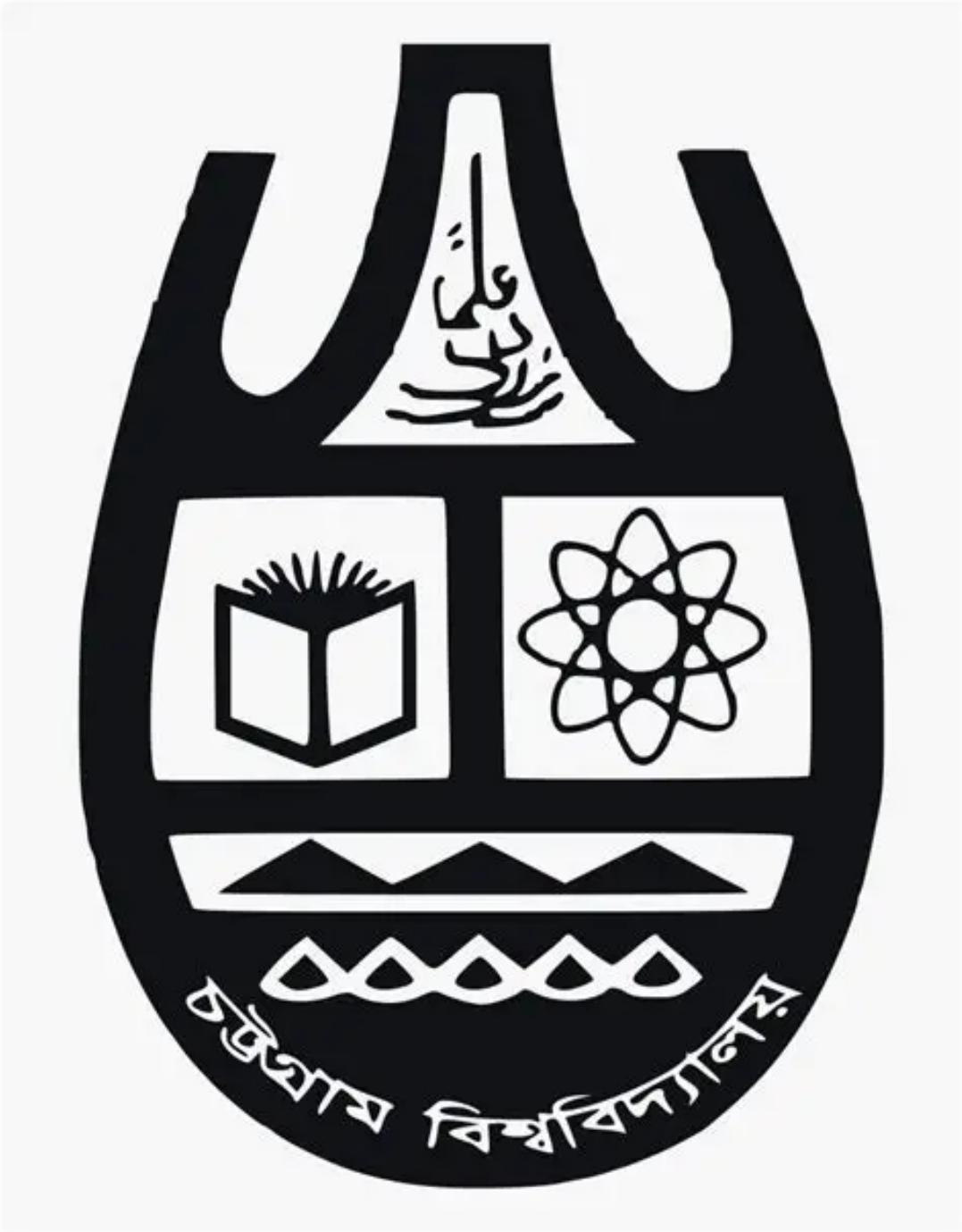
- Dealer(dealer_id, name, address, phone)
- Sales_Staff(staff_id, dealer_id, name, phone)
- Customer(customer_id, name, address, phone, email)
- Sales_Order(order_id, customer_id, dealer_id, staff_id, VIN, order_date, agreed_price, order_status)

Relationships:

- Brand has many Models
- Model offers many Options
- Vehicle is an instance of one Model
- Vehicle may have many selected Options
- Dealer inventories Vehicles
- Dealer employs Sales_Staff
- Customer places Sales_Orders
- Sales_Staff handles Sales_Orders

Question 13
Analyze, solve, and present the solution of Problem 6.23 (ER diagram only).

Answer:



E-R diagram for Problem 6.23

A suitable design includes:

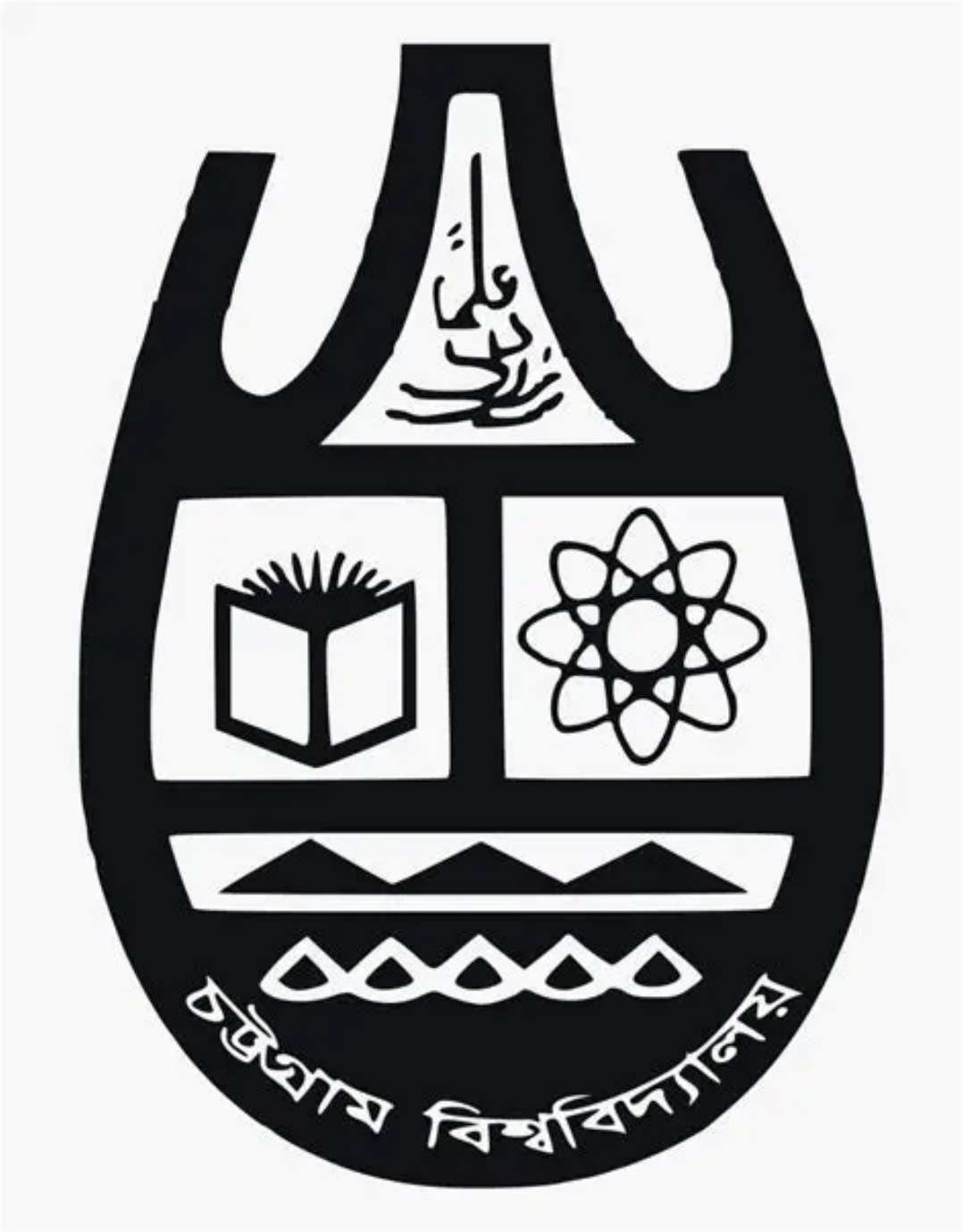
- Customer(customer_id, name, address, phone, email)
- Shipment(shipment_id, sender_id, receiver_id, ship_date, service_type)
- Package(tracking_no, shipment_id, weight, dimensions, status)
- Location(location_id, location_type, description)

- Truck(location_id, registration_no, route_name)
- Plane(location_id, tail_no, airline_code)
- Airport(location_id, airport_code, city, country)
- Warehouse(location_id, warehouse_code, address)
- Package_Tracking(tracking_event_id, tracking_no, location_id, event_time, event_status)

Key idea: the current location of a package is the most recent tracking record.

Question 14
Analyze, solve, and present the solution of Problem 6.24 (ER diagram only).

Answer:



E-R diagram for Problem 6.24

A suitable airline database design includes:

- Airport(airport_code, name, city, country)
- Aircraft(aircraft_id, model, seat_capacity)
- Flight_Route(route_id, flight_no, origin_code, destination_code, scheduled_duration)
- Flight_Instance(instance_id, route_id, aircraft_id, departure_time, arrival_time, status)

- Passenger(passenger_id, name, phone, email, passport_no)
- Booking(booking_id, passenger_id, instance_id, seat_no, booking_class, booking_status)
- Crew_Member(crew_id, name, role)
- Crew_Assignment(crew_id, instance_id, duty)

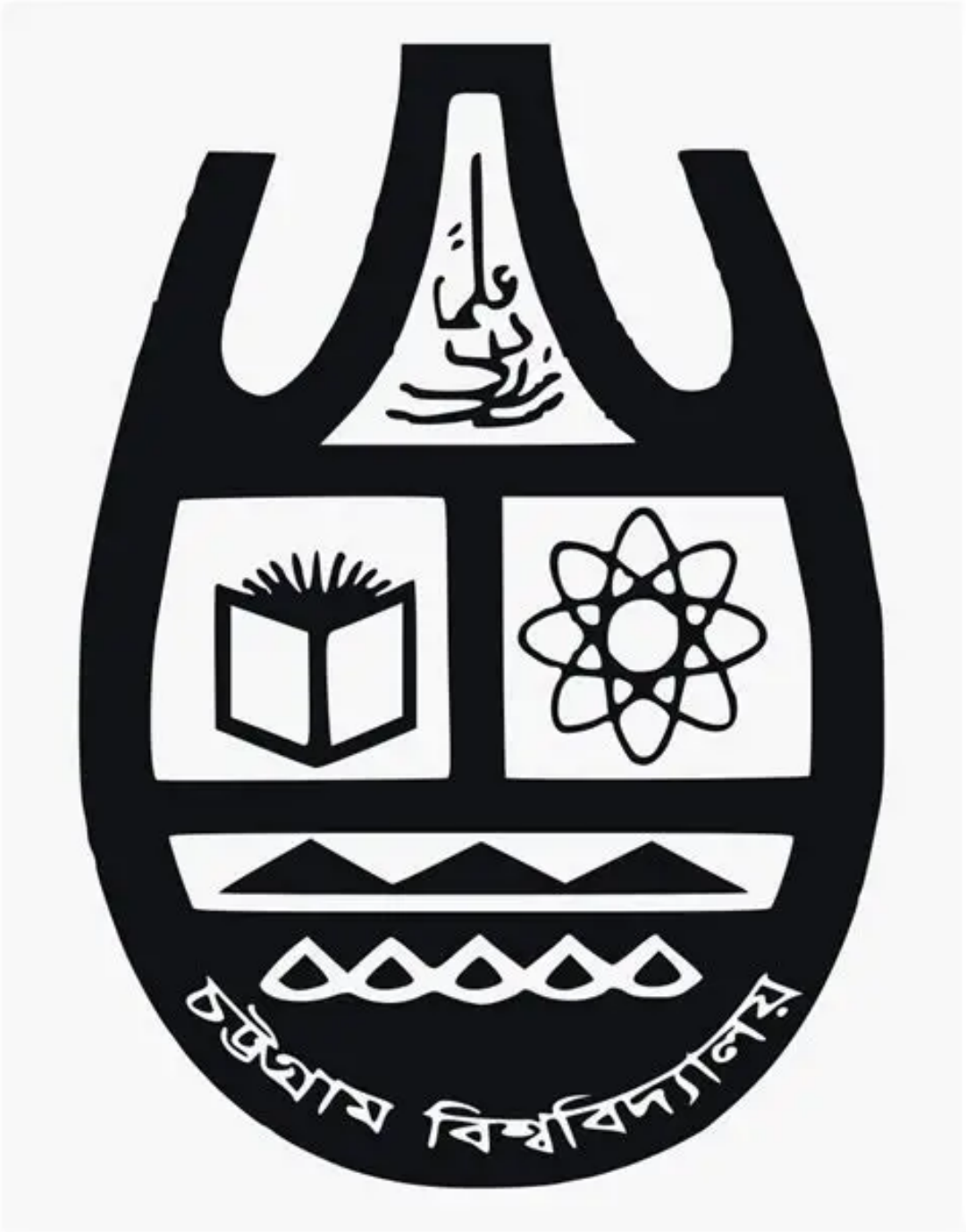
Relationships:

- A flight route has an origin airport and a destination airport
- A flight instance is a scheduled occurrence of a flight route
- An aircraft is assigned to a flight instance
- A passenger books a flight instance
- Crew members are assigned to flight instances

Question 15

Analyze, solve, and present the solution of Problem 6.26.

Answer:



Specialization hierarchy for Problem 6.26

A suitable generalization/specialization hierarchy is:

Motor_Vehicle → Motorcycle, Passenger_Car, Van, Bus

Superclass attributes:

- VIN
- brand
- model
- model_year
- color
- engine_no
- fuel_type
- price

Subclass-specific attributes:

Motorcycle:

- engine_capacity_cc
- bike_type
- has_sidecar

Passenger_Car:

- number_of_doors
- trunk_capacity
- body_style

Van:

- cargo_volume
- payload_capacity
- sliding_door_flag

Bus:

- seating_capacity
- standing_capacity
- route_service_type

This placement is appropriate because common attributes stay in the superclass while specialized attributes remain in the subclasses.